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Supervised Learning

Start easy

Learn through in/out pairs

❖ Tabular

- Classification
- Regression

❖ Temporal

- Survival Analysis

The bad guys

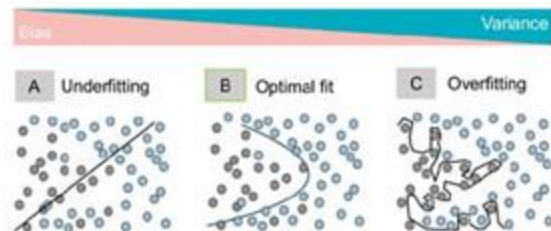
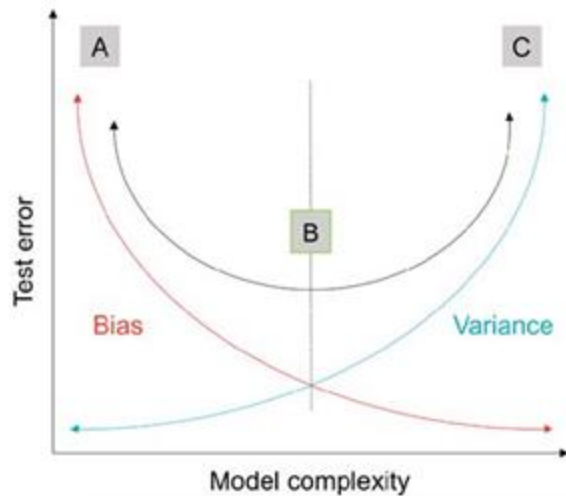
Where's the train error?

❖ Bias*

- Your model is dumb (underfit)
- Solution: More complexity/features

❖ Variance

- Your model is arrogant *and* dumb (overfit)
- Solution: Less complexity/features





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Regression/Classification

Labeling things

Generalizing is **wrong** (but too often it works)

- ❖ The key to AI



- ❖ $a \rightarrow x, e \rightarrow x, i \rightarrow x, o \rightarrow x$ **THUS** $u \rightarrow x$

- ❖ how wrong you are depends on the distribution



Generalizing is **wrong** (but too often it works)

❖ Beyond inductive/supervised

Plato->Mortal, Cicero->Mortal **THUS** Socrates->Mortal

■ Deductive (programming)

- Humans -> Mortal, Socrates -> Human **THUS** Socrates->Mortal

■ Abductive (generative models)

- Socrates -> Mortal, Human-> Mortal **THUS** Socrates->Human



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Regression

A must baseline (with ME/SM permission)

Linear Regression

- ❖ Relation between indep. & dep. vars (continuous)

$$y = b_0 + b_1 * x$$

- ❖ Mean Sq. Error

- ❖ Assume linearity (check with scatter!)

$$MSE = \frac{1}{n} \sum \left(\underbrace{y - \hat{y}}_{\substack{\text{The square of the difference} \\ \text{between actual and} \\ \text{predicted}}} \right)^2$$

- ❖ Sensitive to outliers

- Assume Gaussian residuals (squared)

Linear Regression example

$$y = b_0 + b_1 * x$$

❖ Make a scatter

❖ Estimate intercept (zero)

❖ Estimate slope

Hours Studied (X)	Final Exam
Score (Y)	
2	68
3	66
4	80
5	83
6	96
7	92

Logistic regression (classifier)

- ❖ Probability of belonging to a class (categorical)
- ❖ Logarithmic (log) loss (aka cross entropy loss)
- ❖ Assume linearity
- ❖ Sensitive to outliers

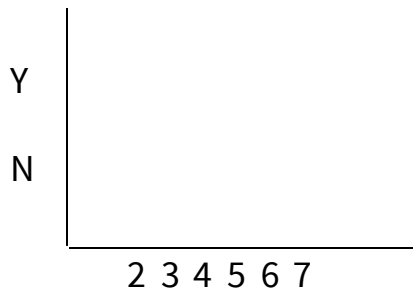
- Assume Gaussian residuals

$$L = -\frac{1}{m} \sum_{i=1}^m y_i \cdot \log(\hat{y}_i)$$

Logistic Regression

$$L = -\frac{1}{m} \sum_{i=1}^m y_i \cdot \log(\hat{y}_i)$$

- ❖ Make a scatter
- ❖ Fit a sigmoid



Hours Studied (X)

2

3

4

5

6

7

Pass Exam (Y)

N

Y

N

N

Y

Y

Other regressions

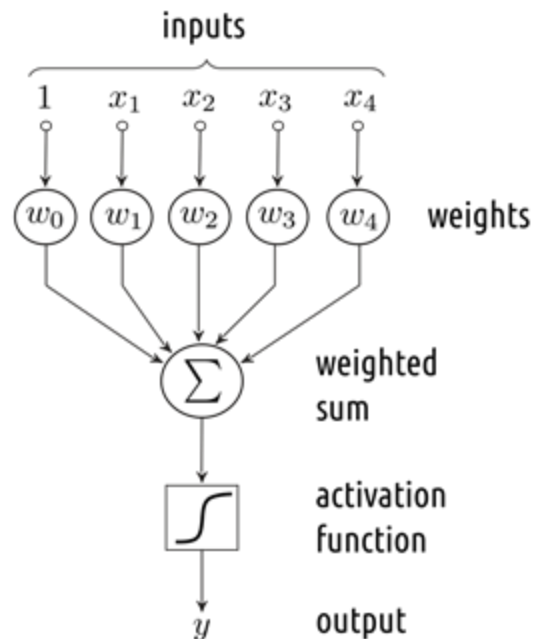
❖ Multi-class

- Multinomial LR
- One set of params per class + *softmax*

$$y_{c1} = b_0 + b_1 * x_1 + b_2 * x_2 + \dots$$

❖ Ordered

- Ordinal LR





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Classification

It's all about the buckets

Classification

❖ The line in the sand

Classical Classify

- ❖ $\text{diff}(\text{set}(A), \text{set}(B))$ aka boundary
- ❖ Binary / Multiclass
- ❖ Tabular data (i.i.d.)
 - Att & feat
 - Cat/Num & a special one
 - Matrix or vectors (with tuple)

Binary Classification Evaluation

		Predicted condition	
		Positive (PP)	Negative (PN)
Actual condition	Positive (P)	True positive (TP), hit	False negative (FN), type II error, miss, underestimation
	Negative (N)	False positive (FP), type I error, false alarm, overestimation	True negative (TN), correct rejection

❖ Top of the line

- Accuracy
 - Top-x
- Precision
- Recall
- F1-score

$$TP+TN/P+N$$

$$TP/TP+FP$$

$$TP/TP+FN$$

$$2*((P*R)/(P+R))$$

❖ Curves

- ROC
- PR
- AUC

❖ Confusion Matrices [\[11\]](#)

KNN: The algorithm

- ❖ Choose K
- ❖ For each sample
 - Calculate distances to others
 - Find k -closest
 - Vote

KNN properties

- ❖ Classification
 - Local approximation
 - Non-parametric (don't mind the distribution)
- ❖ Distance definition
 - Scaling/Weighting
- ❖ Feature selection
 - Curse of dim.
 - Noise: Irrelevant/redundant features
- ❖ Robust to outliers

❖ Can be

- A regressor (avg.)
- Weighted by proximity (e.g., $1/d$)
- Simple: No train. Only inference.
- Better with more data
- A minimal baseline (you can probably do better)
- Rarely the best (local only?)

❖ Hyperparameters

■ k

- Noisy if small. Uniform if large
- High impact
- Dynamic w/ data size
- Mind imbalanced sparsity (weighting helps)

KNN

- ❖ Define distance
 - Scaling?
 - Weighting?
- ❖ Compute distance matrix
- ❖ Predict
 - 150 | 6 | ?
 - 170 | 5 | ?

Weight (g)		Sweetness	Fruit
140	2		Apple
130	3		Apple
165	7		Apple
170	8		Apple
180	9		Orange
160	8		Orange
175	7		Orange
190	6		Orange

Scaling & Normalizing

- ❖ z-score (standardization)

$$z = \frac{x - \mu}{\sigma}$$

- ❖ min-max scaling

$$x_{scaled} = \frac{x - x_{min}}{x_{max} - x_{min}}$$

Distances

❖ Euclidean

$$dist(d_i, d_j) = \sqrt{\sum_k (a_{i,k} - a_{j,k})^2}$$

❖ Weighted Euclidean

$$dist(d_i, d_j) = \sqrt{\sum_k w_k (a_{i,k} - a_{j,k})^2}$$

❖ Cosine

$$cos(d_i, d_j) = \frac{\sum_k a_{i,k} a_{j,k}}{\sqrt{\sum_k a_{i,k}^2} \sqrt{\sum_k a_{j,k}^2}}$$

dot product

product of magnitudes

Resilient to

- High dimensionality
- Changes in magnitude